

# Chapter 1

## All the way to the Jordan Form

### 1.1 Preamble

Our goal in this Chapter is to arrive at the Jordan form of a linear transformation over an arbitrary field. Chances are that some of you have already seen the Jordan form of a matrix, most likely over the complex field. You will then appreciate the derivation of it over an arbitrary field, since it is obtained just about in the same way! At a high level, this is a consequence of the fact that elementary matrix operations keep us on the field we work in.

Still, to obtain the Jordan form, we will need to introduce the key concepts of eigenvalues and eigenvectors. No doubt we have all encountered these before, and we are quite used to define eigenvalues from the characteristic equation

$$\lambda: \det(A - \lambda I) = 0 \rightarrow \text{eigenvalues}$$

and associate eigenvector(s) to each eigenvalue, so that each eigenvalue has an associated algebraic and geometric multiplicity. Of course, this is absolutely fine, but for the time being we will pretend that we do not know this (yet).

If not the Jordan form itself, you have probably seen already the following prototypical result of *diagonalization* of matrices with real or complex entries by real valued or complex valued matrices, respectively:

$$\text{“Given } A \in \begin{cases} \mathbb{R}^{n \times n}, & \text{with distinct real eigenvalues} \\ \mathbb{C}^{n \times n}, & \text{with distinct eigenvalues} \end{cases},$$

then  $A$  is diagonalizable over the real, respectively complex, field.”

Indeed, quite frequently, in undergraduate courses in Linear Algebra, this is the result which is presented, leaving the misleading expectation that all matrices are diagonalizable! Still, let us stress the key assumption spelled out above, and whose relevance will soon be clarified: Matrices were assumed to have distinct eigenvalues in the same field as their entries. E.g., matrices with real entries were assumed to have real eigenvalues.

- In this chapter, however, we want to keep ourselves somewhat disconnected from the specific numerical type of the *entries* in our matrices. With essentially no major effort, we will be able to obtain results which are valid over an arbitrary field. To do this, we need to brush up on some abstraction.

Our first goal is to introduce the concept of **vector space  $V$  over a field  $\mathbb{F}$**  ( $\mathbb{F}$  will always indicate a general field, it is “where the numbers are”).

## 1.2 Field

A field is a “commutative division ring”.

1. *Ring* (associative, always). It is a set  $R$  with two operations “+” and “ $\cdot$ ” such that  $\forall a, b, c \in R$ 
  - (1)  $a + b \in R$
  - (2)  $a + b = b + a$
  - (3)  $(a + b) + c = a + (b + c)$
  - (4)  $\exists 0 \in R : a + 0 = a$  (zero element)
  - (5)  $\exists -a \in R : a + (-a) = 0$  (opposite element)
  - (6)  $a \cdot b \in R$
  - (7)  $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
  - (8)  $a \cdot (b + c) = a \cdot b + a \cdot c$ , and  $(a + b) \cdot c = a \cdot c + b \cdot c$
2. Moreover, if  $a \cdot b = b \cdot a$ , then it is called a *commutative ring*.
3. *Division ring*: the nonzero elements form a group under multiplication.  
[NB: This implies that there is a *unit* element “1” in the ring and every non-zero element  $a$  has an inverse, write it  $1/a$ .]

**Examples 1.2.1** *Examples of fields.*

- *Familiar fields we work with:  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{Q}$  (with the usual + and  $\cdot$ ). These are fields with infinitely many elements.*
- *$\mathbb{Z}$  (usual + and  $\cdot$ ) is a commutative ring, it has unit element, but it is not a field ( $\nexists$  “inverse” under multiplication).*

- $R = \{\mathbb{Z} \pmod{7}\} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$  is a finite field.  
(NB: if we had taken, say,  $\{\mathbb{Z} \pmod{6}\}$  then we would have not obtained a field.)

**Recall.** Remember that we say that a field  $\mathbb{F}$  is of “characteristic 0” if it holds that  $na \neq 0$ , for any  $a \in \mathbb{F}$ ,  $a \neq 0$ , and  $n > 0$ , any integer. Here,  $na = \underbrace{a + a + \cdots + a}_{n \text{ times}}$ . Each of  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{Q}$ , is of characteristic 0. On the other hand,  $\mathbb{F}$  is called of *finite characteristic*  $n$  if there exist a smallest positive integer  $n$  such that  $na = 0$ , for all  $a \in \mathbb{F}$ . See the Exercises below.

### Exercises 1.2.2

- (1) Give an example of a ring without unit element.
- (2) Explain why  $\{\mathbb{Z} \pmod{7}\}$  is a field, but  $\{\mathbb{Z} \pmod{6}\}$  is not. [Hint: Can one have (or not)  $a \cdot b = 0$  without  $a = 0$  or  $b = 0$ ?]
- (3) Take the field  $\{\mathbb{Z} \pmod{5}\} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ . Complete the addition and multiplication Tables. What is  $\bar{1} - \bar{2}$ ?
- (4) Give an example of a field of characteristic 2.
- (5) Show that if  $\mathbb{F}$  is of finite characteristic  $n$ , then  $n$  is a prime number.

## 1.3 Vector Space

A *vector space*, also called a *linear space*, will be indicated by  $V$ , with associated field  $\mathbb{F}$  ( $V$  contains the vectors,  $\mathbb{F}$  the scalars).

**Agreement 1.** To avoid unneeded complications, we will always restrict to the case in which the elements of the set  $V$  have numerical type from the field  $\mathbb{F}$ .

$V$  is a set ( $\neq \emptyset$ ) with “addition”  $+$  such that with respect to  $+$ ,  $V$  is an Abelian (i.e., commutative) group. That is, for all  $v, w, z \in V$  we have

$$\begin{aligned} v + w &= w + v \in V \\ \exists 0 \in V : v + 0 &= 0 + v = v \\ v + (-v) &= 0 \\ v + (w + z) &= (v + w) + z . \end{aligned}$$

Moreover,  $\forall \alpha \in \mathbb{F}, v \in V \Rightarrow \alpha v \in V$  (multiplication by a scalar) and  $\forall \alpha, \beta \in \mathbb{F}$ :

$$\alpha(v + w) = \alpha v + \alpha w, \quad (\alpha + \beta)v = \alpha v + \beta v, \quad \alpha(\beta v) = \alpha\beta v, \quad 1 v = v .$$

**Exercise 1.3.1 (Mandatory review!)** Refresh yourself with the concepts of linear independence, dimension, basis, subspace, etc., for a vector space  $V$ .

**Agreement 2.**  $V$  will always be **finite dimensional**, the default dimension of  $V$  being the value  $n$ .

One useful result that you have certainly seen before, and that we will use often is the following:

**Theorem 1.3.2 (Completion to a basis)** If  $\{v_1, \dots, v_p\}$  are  $p$  linearly independent vectors in  $V$ , and  $V$  has dimension  $n > p$ , then it is always possible to complete the given set of vectors to a basis for  $V$ . That is, to find vectors  $v_{p+1}, \dots, v_n$ , such that  $\{v_1, \dots, v_n\}$  is a basis for  $V$ .  $\square$

**Example 1.3.3** Examples of familiar vector spaces are:  $V = \mathbb{R}^n$  (with  $\mathbb{F} = \mathbb{R}$ ),  $V = \mathbb{C}^n$  (with  $\mathbb{F} = \mathbb{C}$ ),  $V = \mathbb{Q}^n$  (with  $\mathbb{F} = \mathbb{Q}$ ). Also the set of real valued  $(m, n)$  matrices  $V = \mathbb{R}^{m \times n}$ , etc..

**Definition 1.3.4 (Linear Transformation)** Consider a mapping between vector spaces  $V$  and  $W$  (both over  $\mathbb{F}$ ):

$$T : V \rightarrow W.$$

If

- (1)  $T(v_1 + v_2) = Tv_1 + Tv_2$ , for all  $v_1, v_2 \in V$ ,
- (2)  $T(\alpha v) = \alpha Tv$ , for all  $v \in V$  and  $\alpha \in \mathbb{F}$ ,

then  $T$  is called homomorphism (or linear mapping, or linear transformation).

The set of all linear mappings between  $V$  and  $W$  is written as  $\text{Hom}(V, W)$ .

- Very important are the linear maps of  $V$  into itself,  $T : V \rightarrow V$ . These are elements of  $\text{Hom}(V, V)$ .
- It is simple to see that under the obvious operation of “+”,  $\text{Hom}(V, W)$  is itself a vector space over  $\mathbb{F}$ :

$$(T + S)(v) = Tv + Sv, \quad (\alpha T)(v) = \alpha T(v).$$

**Agreement 3.** We will call two linear mappings  $S$  and  $T$  equal, and write  $S = T$ , if  $S$  and  $T$  agree on a basis of  $V$ .

**Theorem 1.3.5** *Hom(V, W) is a vector space of dimension mn, where  $n = \dim(V)$  and  $m = \dim(W)$ . In particular,  $\dim(\text{Hom}(V, V)) = n^2$ .*

**Pf.** We build a basis of  $\text{Hom}(V, W)$ . Let  $\{v_1, \dots, v_n\}$  be a basis for  $V$ , and  $\{w_1, \dots, w_m\}$  be a basis for  $W$ . For  $i = 1, \dots, m$ ,  $j = 1, \dots, n$ , define the linear transformations

$$T_{ij} : V \rightarrow W \quad \text{s.t. if } v \in V, \quad v = \lambda_1 v_1 + \dots + \lambda_n v_n \Rightarrow T_{ij}v = \lambda_j w_i,$$

that is:

$$T_{ij}v_k = \begin{cases} 0, & k \neq j \\ w_i, & k = j \end{cases}.$$

There are  $mn$  such  $T_{ij}$ . Now we show that they are a basis, that is:

- (a) they span  $\text{Hom}(V, W)$  (over  $\mathbb{F}$ );
- (b) they are linearly independent.

Take any  $S \in \text{Hom}(V, W)$ . Now, for  $i = 1, \dots, n$ ,  $Sw_i \in W \Rightarrow Sw_i = \alpha_{1i}w_1 + \dots + \alpha_{mi}w_m$  ( $\alpha_{ji} \in \mathbb{F}$ ). Also, take  $S_0 = \sum_{k=1}^m \sum_{j=1}^n \alpha_{kj} T_{kj}$ . Then:  $S_0v_i = (\alpha_{11}T_{11} + \dots + \alpha_{m1}T_{m1} + \dots + \alpha_{mn}T_{mn})v_i = \alpha_{1i}T_{1i}v_i + \alpha_{2i}T_{2i}v_i + \dots + \alpha_{mi}T_{mi}v_i = \alpha_{1i}w_1 + \dots + \alpha_{mi}w_m$ .

$\therefore Sw_i = S_0v_i$ ,  $i = 1, \dots, n$ , and  $S$  is arbitrary  $\therefore S$  and  $S_0$  agree on a basis of  $V$  and  $\therefore \{T_{ij}\}$  span  $\text{Hom}(V, W)$ .

Next, we show that the  $\{T_{ij}\}$ 's form a linearly independent set. Suppose not. Then, there  $\exists \beta_{ij} \in \mathbb{F}$ ,  $i = 1, \dots, m$ ,  $j = 1, \dots, n$ , not all 0 such that  $\sum_{i,j} \beta_{ij} T_{ij} = 0$ . For any given  $k = 1, \dots, n$ , we then have

$$0 = \left( \sum_{i,j} \beta_{ij} T_{ij} \right) v_k = \beta_{1k} w_1 + \dots + \beta_{mk} w_m,$$

but  $\{w_1, \dots, w_m\}$  are linearly independent, and therefore  $\beta_{1k} = \dots = \beta_{mk} = 0$ , and so, since  $k$  was arbitrary, we get  $\beta_{ij} = 0$  for all  $i, j$ . Contradiction.  $\square$

**Examples 1.3.6**  $V = \mathbb{R}^n$ ,  $W = \mathbb{R}^m$ ;  $\text{Hom}(V, W) = \{\mathbb{R}^n \rightarrow \mathbb{R}^m\}$  (which will soon be identified with  $\mathbb{R}^{m \times n}$ ) has dimension  $mn$ .  $V = W = \mathbb{C}^n$ ,  $\text{Hom}(V, V) = \{\mathbb{C}^n \rightarrow \mathbb{C}^n\}$  (which will soon be identified with  $\mathbb{C}^{n \times n}$ ) has dimension  $n^2$  (over  $\mathbb{C}$ ).

**Definition 1.3.7** If  $T, S \in \text{Hom}(V, V)$ , then  $TS$  will indicate the composite transformation:

$$TS : V \rightarrow V : v \in V \rightarrow T(Sv) \in V.$$

**Exercise 1.3.8** Let  $T, S \in \text{Hom}(V, V)$ . Verify that  $TS$  is also a linear transformation hence an element of  $\text{Hom}(V, V)$ .

**Remark 1.3.9** *There exists  $I \in \text{Hom}(V, V)$  such that  $IT = TI = T$  (identity element). This is simply because there exists  $I$  such that  $Iv = v$ ,  $\forall v \in V$ ; just take  $\alpha_{ji} = 0$ ,  $j \neq i$ ,  $\alpha_{ii} = 1$  in the construction of Theorem 1.3.5. [In abstract terms, we say that  $\text{Hom}(V, V)$  is an algebra with unit element.]*

**Definition 1.3.10** *A linear transformation  $T \in \text{Hom}(V, V)$  is invertible –or nonsingular– if there exists  $S \in \text{Hom}(V, V) : TS = ST = I$ . We write it as  $S = T^{-1}$ . If  $T$  is not invertible, it is also called singular.*

### 1.3.1 Minimal Polynomial

Let  $T \in \text{Hom}(V, V)$ . Below,  $T^k$  indicates the repeated action of  $T$ ,  $k$  times:  $T^k(\cdot) = T(T \cdots (T(\cdot)) \cdots)$  ( $k$ -times). For completeness, we also define  $T^0 = I$ . Note that, see Exercise 1.3.8,  $T^k \in \text{Hom}(V, V)$ , for all  $k = 0, 1, 2, \dots$ .

An important consequence of the fact that  $\text{Hom}(V, V)$  is a vector space is the following.

**Theorem 1.3.11** *Given any element  $T \in \text{Hom}(V, V)$ , there is a nontrivial polynomial  $q(x) \in \mathbb{F}[x]$  of degree at most  $n^2$  such that  $q(T) = 0$ .*

**Pf.** Take  $T \in \text{Hom}(V, V)$ . Form  $I, T, T^2, \dots, T^{n^2}$ . These are  $n^2 + 1$  elements  $\Rightarrow$  they are linearly dependent.

$\Rightarrow \exists \alpha_0, \alpha_1, \dots, \alpha_{n^2} \in \mathbb{F}$  (not all 0) such that  $\alpha_0 I + \alpha_1 T + \dots + \alpha_{n^2} T^{n^2} = 0$ ; that is,  $T$  annihilates the nontrivial polynomial  $q(x) = \alpha_0 + \alpha_1 x + \dots + \alpha_{n^2} x^{n^2}$ .  $\square$

**Definition 1.3.12** *The monic<sup>1</sup> polynomial of lowest degree  $p(x)$  such that  $p(T) = 0$  is called minimal polynomial for  $T$  (over  $\mathbb{F}$ ).*

**Exercises 1.3.13** *Below,  $T \in \text{Hom}(V, V)$  and  $V$  is  $n$ -dimensional, unless otherwise stated.*

- (1) *Show that the minimal polynomial of  $T$  is unique. [Note that there must exist one, since  $V$  is finite dimensional.]*
- (2) *Show that  $T \in \text{Hom}(V, V)$  is invertible  $\Leftrightarrow$  the constant term of the minimal polynomial is not 0. [Hint: Seek directly a form for  $T^{-1}$ .]*
- (3) *Show that if  $T$  annihilates a polynomial  $q(x) \in \mathbb{F}[x]$ , and  $S$  is invertible, then also  $S^{-1}TS$  annihilates  $q(x)$ . In particular,  $T$  and  $S^{-1}TS$  have same minimal polynomial.*

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<sup>1</sup>The leading coefficient is  $1 \in \mathbb{F}$

- (4) Show that  $T$  singular  $\Leftrightarrow \exists v \in V, v \neq 0 : Tv = 0$ .
- (5) This is about nonsingular linear maps.
- (a) Prove that the nonsingular elements in  $\text{Hom}(V, V)$  form a group.
- (b) Let  $\mathbb{F} = \{\mathbb{Z} \pmod{2}\}$ , and let  $V$  be 2-dimensional over  $\mathbb{F}$ . Compute the group of nonsingular elements in  $\text{Hom}(V, V)$ .
- (6) The transformation  $T \in \text{Hom}(V, V)$  is called nilpotent, if  $T^k = 0$ , for some (integer)  $k > 0$ .
- (a) Show that if  $T$  is nilpotent and  $Tv = \alpha v$ , for some  $v \in V, v \neq 0$ , then  $\alpha = 0$ .
- (b) Show that if  $T$  is nilpotent and  $\alpha_0 \neq 0$ , then  $S = \alpha_0 I + \alpha_1 T + \cdots + \alpha_p T^p$  is invertible (here,  $p$  is any positive integer).

## 1.4 Eigenvalues and eigenvectors: 1st attempt

Here, let  $T \in \text{Hom}(V, V)$  and  $V$  be  $n$ -dimensional. Because of Exercise 1.3.13-(4), the following is meaningful.

**Definition 1.4.1** A value  $\lambda \in \mathbb{F}$ , if any<sup>2</sup>, for which  $T - \lambda I$  is singular, is called **eigenvalue** of  $T$ .

If such  $\lambda \in \mathbb{F}$  exists, then a vector  $v \in V, v \neq 0$ , for which  $Tv = \lambda v$ , is called **eigenvector** of  $T$  associated to the eigenvalue  $\lambda \in \mathbb{F}$ .

### Exercises 1.4.2

- (1)  $\lambda \in \mathbb{F}$  is an eigenvalue of  $T \Leftrightarrow$  for some  $v \neq 0, v \in V, Tv = \lambda v$ . In particular, 0 is an eigenvalue of  $T$  if and only if  $T$  is singular.
- (2) Show that if  $\lambda \in \mathbb{F}$  is eigenvalue of  $T \in \text{Hom}(V, V)$ , then  $\lambda$  is a root of the minimal polynomial. [Hint: You know that  $q(T) = 0$ .]
- NB:** As a consequence of this exercise,  $T$  has finitely many eigenvalues in  $\mathbb{F}$ .
- (3) If  $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{F}$  are distinct eigenvalues of  $T \in \text{Hom}(V, V)$  and  $v_1, v_2, \dots, v_k$  are the associated eigenvectors, then  $v_1, v_2, \dots, v_k$  are linearly independent.

A consequence of this last exercise is that:

“If  $\dim(V) = n \Rightarrow T$  has at most  $n$  distinct eigenvalues in  $\mathbb{F}$ ”.

(Any  $k$  distinct eigenvalues correspond to  $k$  linearly independent eigenvectors. But  $V$  is  $n$ -dimensional.)

Further, this last fact can be also reformulated as saying:

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<sup>2</sup>Depending on  $\mathbb{F}$ , there may be no  $\lambda \in \mathbb{F}$  for which this occurs

**Theorem 1.4.3 (Basis of Eigenvectors)** *If  $\dim(V) = n$  and  $T$  has  $n$  distinct eigenvalues in  $\mathbb{F} \Rightarrow V$  has a basis of eigenvectors of  $T$ .*  $\square$

**Remark 1.4.4** *We stress that, depending on what  $\mathbb{F}$  is, for a transformation  $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$  there may be no value  $\lambda \in \mathbb{F}$  such that  $T - \lambda I$  is singular. For example, consider the linear transformation  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$  given by  $x_1 \rightarrow 2x_1 + x_2$  and  $x_2 \rightarrow -x_1 + 2x_2$ .*

## 1.5 Matrices and canonical forms

To a linear transformation there is associated a **matrix**; when  $\mathbb{F}$  has infinitely many elements, there are infinitely many matrices. Let us see how this correspondence works.

Consider  $T \in \text{Hom}(V, W)$ , both  $V$  and  $W$  vector spaces over the field  $\mathbb{F}$ . Let  $V$  be  $n$ -dimensional with basis  $\{v_1, \dots, v_n\}$ , and  $W$  be  $m$ -dimensional with basis  $\{w_1, \dots, w_m\}$ . Then, we have

$$Tv_j = \sum_{i=1}^m a_{ij} w_i, \quad j = 1, \dots, n,$$

for some values  $a_{ij} \in \mathbb{F}$ . Thus, we have a collection of  $mn$  values  $a_{ij}$ 's and we associate to  $T$  the matrix  $A = (a_{ij})_{i=1:m, j=1:n} \in \mathbb{F}^{m \times n}$ . In the particular case of  $T \in \text{Hom}(V, V)$ , we obtain a *square* matrix  $A = (a_{ij})_{i=1:n, j=1:n} \in \mathbb{F}^{n \times n}$  and

$$Tv_j = a_{1j}v_1 + \dots + a_{nj}v_n, \quad j = 1, \dots, n.$$

**Example 1.5.1** *Take  $I \in \text{Hom}(V, V)$  (identity transformation), and let  $\{v_1, \dots, v_n\}$  be a basis for  $V$ . Then,  $Iv_j = v_j$ ,  $j = 1, \dots, n$ . Thus, the matrix representation of the identity transformation is the **identity matrix**, which we will also call  $I$ :  $I =$*

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}.$$

*Observe that the matrix representation of the identity transformation is independent of the basis chosen for  $V$ .*

**Exercise 1.5.2** *Refresh yourself with the way matrix product works, and note that the usual definition of matrix product applies in a general field  $\mathbb{F}$ .*

**Exercise 1.5.3** Let  $\{e_i^{(n)}\}_{i=1,\dots,n}$  and  $\{e_i^{(m)}\}_{i=1,\dots,m}$  be the standard bases for  $\mathbb{R}^n$  and  $\mathbb{R}^m$ , respectively. (As usual, these are the standard unit vectors: their  $j$ -th entry is  $(e_i^{(n,m)})_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$ , the only difference being how many entries there are in  $e_i^{(n)}$  and  $e_i^{(m)}$ .)

Find the basis  $\{T_{ij}\}$  of Theorem 1.3.5 for  $\text{Hom}(V, W)$  (equivalently, for  $\mathbb{R}^{m \times n}$ ).

**Exercise 1.5.4** Let  $T, S \in \text{Hom}(V, V)$  and let  $\mathcal{V} = \{v_1, \dots, v_n\}$  be a basis for  $V$  so that  $T$  and  $S$  have matrix representations  $A$  and  $B$ , respectively, with respect to  $\mathcal{V}$ . Show that the composite transformation  $TS : V \rightarrow V$  has matrix representation given by the product  $AB$ .

**Definition 1.5.5** Given a matrix  $A \in \mathbb{F}^{n \times n}$ , we say that  $A$  is invertible if there exists a matrix  $B \in \mathbb{F}^{n \times n}$  such that  $AB = BA = I$ . We write  $B = A^{-1}$  for the inverse of  $A$ .

**Exercise 1.5.6** Show that if a matrix  $A$  is invertible, then the “inverse is unique”.

**Remark 1.5.7** Suppose  $T \in \text{Hom}(V, V)$ . We stress that, in general, the matrix associated to  $T$  depends on the basis chosen for  $V$ . So, while the transformation viewed as mapping from  $V$  to  $V$ , say, is not ambiguous, the matrix representation of  $T$  may be.

The last remark is actually deeper than it sounds, because it suggests to:

*Choose appropriate bases to get nice matrices!*

What precisely is meant by “nice” is in the eye of the beholder, but let us tentatively agree that a nice matrix is one which has a form which reveals important information, such as the eigenvalues, or the rank, etc..

As a matter of fact, this innocent sounding consideration above is central to much linear algebra: we usually (for convenience, simplicity, or necessity) use simple bases to express linear transformations, and end up with matrices which are not necessarily revealing structural information on the transformation itself. Afterwards, we try to transform these matrices into a form which reveals structural information on the transformation. The most important operations we do (see Theorem 1.5.10) are tantamount to changing bases. If we were able to guess the right basis before associating a matrix to the transformation, there would be little scope for further work!

**Example 1.5.8** For example, a nice matrix  $A$ , when  $T$  has  $n$  distinct eigenvalues in  $\mathbb{F}$ , will be obtained upon realizing that relative to its eigenvectors we have (see

Theorem 1.4.3):

$$\text{For } i = 1, \dots, n: Tv_i = \lambda_i v_i \Rightarrow \text{ in this basis } A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}.$$

A matrix  $A$  of this form is called *diagonal*, which means that the only possibly nonzero entries are along the diagonal of  $A$ . We will consider this diagonal form very nice indeed. As a matter of fact, it is tempting to say that a nice matrix is one which is as close as possible to being diagonal. Also, **note** that if  $V$  has a basis of eigenvectors of  $T$ , then in this basis  $T$  is diagonal (regardless of the eigenvalues being distinct).

- A very important result is that matrices representing the same linear transformation in different bases are *similar* matrices.

**Definition 1.5.9** Let  $A, B \in \mathbb{F}^{n \times n}$ . If there exists an invertible matrix  $C \in \mathbb{F}^{n \times n}$ , such that  $B = C^{-1}AC$ , then  $A$  and  $B$  are called *similar matrices*. The relation  $B = C^{-1}AC$  is called a *similarity transformation* of  $A$  and  $B$ .

We have the following fundamental result.

**Theorem 1.5.10** Let  $T \in \text{Hom}(V, V)$ , and let  $A$  be the matrix associated to  $T$  in the basis  $\mathcal{V} = \{v_1, \dots, v_n\}$ , and  $B$  be the matrix associated to  $T$  in the basis  $\mathcal{W} = \{w_1, \dots, w_n\}$ . Then, there exists  $C \in \mathbb{F}^{n \times n}$ , invertible, such that  $B = C^{-1}AC$ . The matrix  $C$  is the matrix associated to the linear transformation  $S$  which changes bases:  $w_j = \sum_{i=1}^n c_{ij}v_i$ ,  $j = 1, \dots, n$ .

**Example 1.5.11** Let  $V = \mathbb{R}^2$  (or also  $\mathbb{Q}^2$ ) and consider the transformation  $T : \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x + 2y \\ 3x - y \end{pmatrix}$ . With respect to the standard basis  $\mathcal{V} = \{e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}\}$ , the matrix representation of  $T$  is  $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ . With the respect to the basis  $\mathcal{W} = \{w_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, w_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\}$ , the matrix associated to  $T$  is given by  $B = \begin{bmatrix} -5/2 & 1/2 \\ 3/2 & 5/2 \end{bmatrix}$ . In this example, the matrix  $C$  of Theorem 1.5.10 giving  $C^{-1}AC = B$  is  $C = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ .

**Exercises 1.5.12**

- (1) Prove Theorem 1.5.10.  
 (2) Show that similarity is an equivalence relation, and that it does not change the rank of a matrix associated to  $T$ .

**Exercises 1.5.13** Recall that two matrices  $A, B \in \mathbb{F}^{n \times n}$  commute, by definition, if  $AB = BA$ .

- (1) Show that the only matrices in  $\mathbb{F}^{n \times n}$  commuting with all matrices in  $\mathbb{F}^{n \times n}$  are  $\alpha I$ ,  $\alpha \in \mathbb{F}$ .  
 (2) Show that there cannot be  $A, B \in \mathbb{R}^{n \times n} : AB - BA = I$ . [Hint: Look at  $\sum_i ((AB)_{ii} - (BA)_{ii})$ .]  
 (3) Let  $A \in \mathbb{F}^{n \times n}$ , and suppose  $A^k = 0$  for some  $k$ . Show that  $I + A$  is invertible and give a compact expression for  $(I + A)^{-1}$ . [Hint: See Exercise 1.3.13-(6)].

- (4) Let  $A \in \mathbb{F}^{n \times n}$ ,  $A = \begin{pmatrix} \alpha & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ 0 & & & \alpha \end{pmatrix}$ ,  $\alpha \neq 0$ . Show that  $A$  is invertible and express its inverse in a compact way. [Hint: write  $A = \alpha I + B$ .]

**Reminder 1.5.14** This is a refresher of frequent matrix manipulations.

- *Block multiplication.* We often multiply matrices partitioned in compatible blocks. For example, let  $A_{ii}, B_{ii} \in \mathbb{F}^{n_i, n_i}$ ,  $i = 1, 2$ , and let  $A_{12}, B_{12} \in \mathbb{F}^{n_1, n_2}$  and  $A_{21}, B_{21} \in \mathbb{F}^{n_2, n_1}$ , then:

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

- *Transpose.* If  $A = (a_{ij})_{i=1:m, j=1:n} \in \mathbb{F}^{m \times n}$ , then the transpose of  $A$  is written  $A^T \in \mathbb{F}^{n \times m}$  and it is given by  $A^T = (a_{ji})_{j=1:n, i=1:m}$ .

### 1.5.1 Determinant, characteristic polynomial, eigenvalues and eigenvectors

Consider the following familiar problem: “Given  $A \in \mathbb{F}^{n \times n}$  and  $b \in \mathbb{F}^n$ , when is there a unique solution  $x \in \mathbb{F}^n$  to the system  $Ax = b$ ”?

Undoubtedly, for the case of  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{F} = \mathbb{C}$ , we are familiar with many equivalent answers to this question, and are also familiar with techniques to find the solution

(when it exists). What we want to appreciate now is that there is nothing new in the case of an arbitrary field  $\mathbb{F}$ .

**Definition 1.5.15** Let  $A \in \mathbb{F}^{n \times n}$ . We define the determinant of  $A$ , written  $\det A$  as the number in  $\mathbb{F}$  given by:

$$\det A = \sum_{\sigma \in S} (-1)^\sigma a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

where  $S$  is the set of permutations on  $\{1, 2, \dots, n\}$ . Here,  $\sigma$  is a permutation, and

$$(-1)^\sigma = \begin{cases} 1, & \text{for even permutations} \\ -1, & \text{for odd permutations} \end{cases}$$

(“even” means that the minimal number of pairwise exchanges, transpositions, needed to produce  $\sigma$  from  $\{1, 2, \dots, n\}$  is even).

Alternatively, one may use the equivalent definition based on the Laplace Expansion formula:

$$\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det A_{ij} = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det A_{ij} , \quad (1.5.1)$$

where  $A_{ij} \in \mathbb{F}^{n-1, n-1}$  is the matrix obtained by deleting the  $i$ -th row and  $j$ -th column of  $A$ , and  $\det a = a$  for scalar  $a \in \mathbb{F}$ .

The values

$$C_{ij} = (-1)^{i+j} \det A_{ij} , \quad i, j = 1, \dots, n, \quad (1.5.2)$$

are called cofactors of  $A$ . In particular, we can write  $\det A = \sum_{i=1}^n a_{ij} C_{ij}$ .

Note that the value  $\det A$  is always in  $\mathbb{F}$ , stated. Also, note that  $\det A = \det A^T$  (just use (1.5.1)). Some other very useful properties which you can verify (see Exercise 1.5.16-(5) below) are:

$$\det I = 1 , \quad \det(AB) = \det(A) \det(B) .$$

### Exercises 1.5.16

- (1) Show that if a column or a row of  $A$  is made up by all 0's, then  $\det A = 0$ .
- (2) Show that the determinant is a multilinear function of the columns of the matrix; that is, it is separately linear in each column. [Hint: All you need to show is that

if  $A \in \mathbb{F}^{n \times n}$  is partitioned column-wise:  $A = [a_1, a_2, \dots, a_n]$ ,  $\alpha \in \mathbb{F}$ , and  $b \in \mathbb{F}^n$ , then  $\det[a_1, \dots, a_i + \alpha b, \dots, a_n] = \det A + \alpha \det[a_1, \dots, b, \dots, a_n]$ .

- (3) Let the matrix  $B$  be obtained from  $A$  by exchanging two columns/rows of  $A$ . Show that  $\det B = -\det A$ . In particular, conclude that if  $A$  has two equal columns/rows, then  $\det A = 0$ .
- (4) Show that if any column/row of  $A$  is a linear combination of any number of the other columns/rows, then  $\det A = 0$ .
- (5) [Cauchy-Binet formula] Let  $A, B \in \mathbb{F}^{n \times n}$ . By directly using Definition 1.5.15, show that

$$\det(AB) = \det(A) \det(B) . \quad (1.5.3)$$

**Lemma 1.5.17 (Cramer's rule)** Let  $A \in \mathbb{F}^{n \times n}$  and  $b \in \mathbb{F}^n$ . A necessary condition for  $x \in \mathbb{F}^n$  to be a solution of  $Ax = b$  is that the entries of  $x$  satisfy the relation

$$x_j \det A = \det A_j , \quad j = 1, \dots, n , \quad (1.5.4)$$

where  $A_j$  is the matrix obtained replacing the  $j$ -th column of  $A$  with  $b$ .

**Pf.** For any  $j = 1, \dots, n$ , take the matrix  $C_j \in \mathbb{F}^{1 \times n}$ :  $C_j = [C_{1j}, \dots, C_{nj}]$ , where these are the  $(i, j)$  cofactors of  $A$ , see (1.5.2). Next, multiply the relation  $Ax = b$  by  $C_j$ :  $C_j Ax = C_j b$ . Writing this out explicitly gives

$$x_1 \sum_{i=1}^n a_{i1} C_{ij} + \dots + x_j \sum_{i=1}^n a_{ij} C_{ij} + \dots + x_n \sum_{i=1}^n a_{in} C_{ij} = \sum_{i=1}^n b_i C_{ij} .$$

Now, the coefficients of  $x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n$ , are all 0 because of Exercise 1.5.16-(4)<sup>3</sup>, the coefficient of  $x_j$  is precisely  $\det A$ , and the right hand side is  $\det A_j$ , which is what we had to prove.  $\square$

An immediate consequence of Lemma 1.5.17 and Exercise 1.5.16-(1) is the following.

**Corollary 1.5.18** If there exists  $v \in \mathbb{F}^n$ ,  $v \neq 0$ , such that  $Av = 0$ , then  $\det A = 0$ .

**Exercise 1.5.19** Show that if  $Ax = b$  does not have a unique solution, then  $\det A = 0$ . (If the field has infinitely many elements, and if there is more than one solution, then there are infinitely many different solutions.)

<sup>3</sup>E.g., observe that  $\sum_{i=1}^n a_{i1} C_{ij}$  is the determinant of the matrix  $A$  with  $j$ -th column replaced by the first, hence this matrix has two equal columns when  $j \neq 1$ .

Next, we have:

**Lemma 1.5.20** *If  $\det A \neq 0$ , then  $A$  is invertible, its inverse,  $A^{-1}$ , is in  $\mathbb{F}^{n \times n}$  and  $\det A^{-1} = \frac{1}{\det A}$ .*

**Pf.** Consider the matrix  $B$  whose  $(i, j)$ -th entry is the  $(j, i)$ -th cofactor of  $A$ :  $B_{ij} = C_{ji}$ ,  $i, j = 1, \dots, n$ .<sup>4</sup>

Now, consider the problem  $AX = I \det A$ , or (written column-wise)

$$A \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & \dots & e_n \end{bmatrix} \det A .$$

So, we have  $Ax_j = e_j \det A$ ,  $j = 1, \dots, n$ , which using Lemma 1.5.17 repeatedly, and the Laplace expansion of the determinant, gives  $(x_j)_k \det A = \det A_k$ , where  $A_k = [a_1, \dots, a_{k-1}, e_j \det A, a_{k+1}, \dots, a_n]$  and  $\det A_k = C_{jk} \det(A)$ , for  $k = 1, \dots, n$ ,  $j = 1, \dots, n$ , from which we get the solution  $X = B$ . Therefore,  $B/\det A$  is the inverse of  $A$ , necessarily unique. By construction, it is in  $\mathbb{F}^{n \times n}$ . The statement about  $\det A^{-1}$  is immediate using (1.5.3).  $\square$

Notice that from Corollary 1.5.18 and Lemma 1.5.20, we get:

**Theorem 1.5.21**  *$A \in \mathbb{F}^{n \times n}$  is invertible  $\iff \det A \neq 0$ .*

An immediate consequence of Theorem 1.5.10, Exercise 1.5.4, the Cauchy-Binet formula (1.5.3), and Lemma 1.5.20 is the following.

**Corollary 1.5.22** *Let  $T \in \text{Hom}(V, V)$ . Then,  $\det T$  is uniquely defined as the determinant of any of its matrix representations.*

With the help of Lemmata 1.5.17 and 1.5.20, we are ready to revisit our definition of eigenvalues and eigenvectors.

**Definition 1.5.23** *Let  $A \in \mathbb{F}^{n \times n}$  and consider the polynomial  $p(\lambda) \in \mathbb{F}[\lambda]$  of degree exactly  $n$  given by*

$$p(\lambda) = \det(\lambda I - A) . \tag{1.5.5}$$

*This is called the characteristic polynomial of  $A$ .*

---

<sup>4</sup>This matrix  $B$  is called the *adjugate*, or *adjunct*, of  $A$ , and it is usually written  $\text{adj} A$

Observe that the roots of  $p(\lambda) = 0$  are unchanged by similarity transformations of  $A$ . Also, observe that –by using the expansion (1.5.1) in (1.5.5)– the characteristic polynomial is a monic polynomial with coefficients in  $\mathbb{F}$ .

Now, through the identification of the linear transformation  $T$  with (any of) its matrix representation  $A$ , and recalling Exercise 1.4.2-(1), we know that 0 is an eigenvalue of  $A$  if and only if  $A$  is singular, and moreover that  $\lambda \in \mathbb{F}$  is an eigenvalue of  $A$  if and only if there is a solution  $v \neq 0$  to the problem  $Av - \lambda v = 0$ . Rewriting this last relation as  $(\lambda I - A)v = 0$ , and using Corollary 1.5.18, this means that  $\lambda$  is a root of  $p(\lambda)$ . Guided by these considerations, we have the following definition.

**Definition 1.5.24 (Eigenvalues and Eigenvectors)** *We define eigenvalues of  $A$  the roots of the characteristic polynomial. To each such root, we associate an eigenvector to be a nonzero solution of the linear system  $Av = \lambda v$ . The pair  $(\lambda, v)$  will be called an eigenpair of  $A$ . The algebraic multiplicity of an eigenvalue  $\lambda$  is the multiplicity of  $\lambda$  as a root of the characteristic polynomial. The geometric multiplicity of  $\lambda$  is the number of linearly independent eigenvectors associated to  $\lambda$ . An eigenvalue is called simple if it has algebraic multiplicity 1, and it is called multiple if it has algebraic multiplicity greater than 1.*

#### Remarks 1.5.25

- (1) **Caution!** *Naturally, we cannot claim that any of the roots of the characteristic polynomial are in  $\mathbb{F}$  (see section 1.5.4). However, we can surely claim that there are no more than  $n$  eigenvalues (counted with multiplicity), in  $\mathbb{F}$  or otherwise.*
- (2) *It is worth emphasizing once more that the eigenvalues are unchanged under similarity of  $A$ . The eigenvectors, instead, are modified by the matrix achieving the similarity transformation: if  $(\lambda, v)$  is an eigenpair of  $A$ ,  $Av = \lambda v$ , and  $B = C^{-1}AC$ , then  $(\lambda, w)$ ,  $w = C^{-1}v$ , is eigenpair of  $B$ .*

## 1.5.2 Diagonalization and Triangularization over $\mathbb{F}$

By virtue of Theorem 1.5.10, we can effectively “identify” linear transformations and matrices. Results for matrices can be equivalently phrased for linear transformations, and viceversa. Definitions of eigenvalues and eigenvectors are likewise transported immediately to the matrix setting, and since they are independent of the matrix representation used for  $T$ , this process is well defined.

**Definition 1.5.26 (Triangular Form)** *A matrix  $R \in \mathbb{F}^{n \times n}$ ,  $R = (r_{i,j})_{i,j=1,\dots,n}$ , is called upper triangular if  $r_{ij} = 0$ ,  $i > j$ . An upper triangular matrix  $R$  with  $r_{ii} =$*

$0, i = 1, \dots, n$ , is called strictly upper triangular. In a perfectly similar way, we define lower triangular:  $L \in \mathbb{F}^{n \times n}$ ,  $L = (l_{i,j})_{i,j=1,\dots,n}$  such that  $l_{ij} = 0$ ,  $i < j$ .

**Definition 1.5.27** We say that a matrix  $A \in \mathbb{F}^{n \times n}$  is diagonalizable over  $\mathbb{F}$  if there exists an invertible  $B \in \mathbb{F}^{n \times n}$  such that  $B^{-1}AB$  is diagonal. Similarly, we say that  $A$  is triangularizable over  $\mathbb{F}$  if there exists an invertible  $B \in \mathbb{F}^{n \times n}$  such that  $B^{-1}AB$  is triangular.

### Diagonal Form.

As a consequence of Theorem 1.4.3, and of the above observations, we have (see the phrase in quotes on page 7):

**Corollary 1.5.28 (First Canonical Form: Diagonalization over  $\mathbb{F}$ )** If  $A \in \mathbb{F}^{n \times n}$

has  $n$  distinct eigenvalues  $\lambda_1, \dots, \lambda_n$  in  $\mathbb{F}$ , then  $AV = V\Lambda$ , where  $\Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}$

and  $V = [v_1, \dots, v_n] \in \mathbb{F}^{n \times n}$  is the matrix of eigenvectors of  $A$ .

**Remark 1.5.29** Corollary 1.5.28 expresses a sufficient condition, but naturally a matrix may be diagonalizable over  $\mathbb{F}$  even if it has repeated eigenvalues, as long as there are  $n$  linearly independent eigenvectors in  $\mathbb{F}^n$ .

**Exercise 1.5.30** Let  $M \in \mathbb{F}^{n \times n}$  be  $M = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$  with  $A \in \mathbb{F}^{n_1 \times n_1}$  and  $B \in \mathbb{F}^{n_2 \times n_2}$ ,  $n_1 + n_2 = n$ . Then, show that  $M$  is diagonalizable (over  $\mathbb{F}$ ) if and only if  $A$  and  $B$  are. [Hint: For the harder part, partition the eigenvectors  $v_i$  of  $M$  as  $v_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}$ , with  $x_i \in \mathbb{F}^{n_1}$ ,  $y_i \in \mathbb{F}^{n_2}$ ,  $i = 1, 2, \dots, n$ , and consider the collection of vectors  $\{x_1, \dots, x_n\}$  and  $\{y_1, \dots, y_n\}$ . What is their rank?]

The next theorem is concerned with the situation when two matrices can be diagonalized by the same similarity transformation. [As it turns out, a similar result holds also for other canonical forms, see Exercise 1.5.41.]

**Theorem 1.5.31 (Simultaneous Diagonalizability)** Let  $A, B \in \mathbb{F}^{n \times n}$  be diagonalizable (over  $\mathbb{F}$ ). Then, they are simultaneously diagonalizable if and only if  $AB = BA$ .

**Pf.** Suppose there exists  $V$  such that  $V^{-1}AV = D_A$  and  $V^{-1}BV = D_B$ , with  $D_A$  and  $D_B$  both diagonal. Then, obviously  $D_AD_B = D_BD_A$  and thus

$$VD_AV^{-1}VD_BV^{-1} = VD_BV^{-1}VD_AV^{-1} \Rightarrow AB = BA.$$

Next, suppose that  $AB = BA$  and let  $V$  be the similarity that diagonalizes  $A$ ,  $V^{-1}AV = D_A$ , ordering the eigenvalues so that equal eigenvalues appear consecutively along the diagonal of  $D_A$ :  $D_A = \text{diag}(\lambda_k I_{n_k}, k = 1, \dots, p)$ . Let  $C = V^{-1}BV$ , so that  $AB = BA$  rewrites as  $D_AC = CD_A$ . Partitioning  $C$  in block form conformally to  $D_A$ 's partitioning, we immediately get that  $C = \text{diag}(C_{kk}, k = 1, \dots, p)$ . Since  $B$  is diagonalizable, so is  $C$ , and thus so are the blocks  $C_{kk}$ ,  $k = 1, \dots, p$  (see Exercise 1.5.30). Let  $T_k$  be the similarities diagonalizing these blocks  $C_{kk}$ ,  $k = 1, \dots, p$ , and let  $T = \text{diag}(T_k, k = 1, \dots, p)$ , so that  $T^{-1}CT$  is diagonal. Now, given the block diagonal structure of  $T$ , we also have  $T^{-1}D_AT = D_A$  and so:

$$T^{-1}(V^{-1}AV)T \quad \text{and} \quad T^{-1}(V^{-1}BV)T$$

are both diagonal. □

Although diagonal form is very nice, it may be impossible to achieve it for a certain transformation. In other words, for a certain given transformation  $T$ , there may be no basis leading to a diagonal matrix representation of  $T$ , even if all eigenvalues are in  $\mathbb{F}$ . The simple example  $A = \begin{bmatrix} 1 & 2 \\ -2 & -3 \end{bmatrix}$  supports this claim. (Both eigenvalues are equal to  $-1$  and there is only one eigenvector).

### Triangular Form.

The next “nice” form we look at is upper *triangular*.

**Fact 1.5.32 (Backward substitution)** *If  $R \in \mathbb{F}^{n \times n}$  is upper triangular and nonsingular, then we can uniquely solve the linear system*

$$Rx = b, \quad \text{for } b \in \mathbb{F}^n,$$

*getting  $x \in \mathbb{F}^n$  one entry at a time, starting with  $x_n$ , then  $x_{n-1}, \dots, x_1$ .*

**Remark 1.5.33** *Fact 1.5.32 above is a very useful result. The algorithm to obtain  $x$  is called “backward substitution.” The key ingredients are nonsingularity and the fact that the arithmetic operations we perform are keeping us in the field.*

**Exercises 1.5.34**

- (1) Develop the backward substitution algorithm. Convince yourself that all arithmetic operations keep you in the field  $\mathbb{F}$ .
- (2) Show that if  $R$  is upper/lower triangular, and it is invertible, then also  $R^{-1}$  is upper/lower triangular.
- (3) Show that if  $R \in \mathbb{F}^{n \times n}$  is triangular (upper or lower), then its eigenvalues are precisely the elements on the diagonal of  $R$ , and only these. Can you show it without using determinants? Can you claim that  $R$  has  $n$  linearly independent eigenvectors?
- (4) Suppose that  $M \in \mathbb{F}^{n \times n}$  is in the block form  $M = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix}$ , where  $A \in \mathbb{F}^{n_1 \times n_1}$  and  $B \in \mathbb{F}^{n_2 \times n_2}$ ,  $n_1 + n_2 = n$ , and that  $M$  has eigenvalues in  $\mathbb{F}$ . Then, show that  $\lambda \in \mathbb{F}$  is an eigenvalue of  $M$  if and only if it is an eigenvalue of either  $A$  or  $B$ , or of both  $A$  and  $B$ .

An important consequence of Exercise 1.5.34-(4) is the following Lemma, which also introduces an important new concept.

**Lemma 1.5.35 (Invariant Subspace Lemma)** *Let  $B \in \mathbb{F}^{n \times n}$ , with all eigenvalues in  $\mathbb{F}$ , and let  $V \in \mathbb{F}^{n \times p}$ ,  $1 \leq p \leq n$ , be a full rank matrix such that*

$$BV = VC, \quad \text{where } C \in \mathbb{F}^{p \times p}. \quad (1.5.6)$$

*Then, these statements hold.*

- (i) *The eigenvalues of  $C$  are also eigenvalues of  $B$ .*
- (ii) *There exists an eigenvector of  $B$  in the column-space of  $V$ .*

**Pf.** To show part (i), extend the columns of  $V$  to a basis of  $\mathbb{F}^n$  by using the columns of a matrix  $W$ , so that the matrix  $T = \begin{bmatrix} V & W \end{bmatrix}$  is invertible. Since  $BV = VC$ , we have  $BT = T \begin{bmatrix} C & D \\ 0 & E \end{bmatrix}$  from which part (i) follows.

To get part (ii), take an eigenpair  $(\mu, z)$  of  $C$ :  $Cz = \mu z$ ,  $z \neq 0$  (and  $\mu \in \mathbb{F}$  because of (i)). Now,  $B(Vz) = \mu(Vz)$ , and since  $V$  is full rank, then  $Vz \in \mathbb{F}^n$  cannot be zero and therefore it is an eigenvector of  $B$  and it belongs to the column space of  $V$ .  $\square$

**Remark 1.5.36** *The relation expressed by (1.5.6) is called an invariant subspace relation, in that the column space of  $V$  is left invariant by the action of  $B$ .*

- The next result tells us that “If  $T \in \text{Hom}(V, V)$  has all eigenvalues in  $\mathbb{F}$ , then there is a basis of  $V$  in which  $T$  is triangular” (i.e., its matrix representation in this basis is). This result is effectively akin to the famous Schur’s form of a matrix. Exercises 1.5.34-(2)&(4) will be used in the proof of the Theorem below.

**Theorem 1.5.37 (Second Canonical Form: Triangularization over  $\mathbb{F}$ )** *Suppose  $A \in \mathbb{F}^{n \times n}$  has all its eigenvalues in  $\mathbb{F}$ . Then, there is an invertible matrix  $B \in \mathbb{F}^{n \times n}$  such that  $R = B^{-1}AB$  is (upper) triangular. Moreover, we can arrange the entries on the diagonal of  $R$  (the eigenvalues) in any way we like.*

**Pf.** Suppose we have an eigenvalue labeled as  $\lambda_1$ , and that this is the first entry we want to see along the diagonal of  $R$ .

The proof is by induction on  $n$ . Let  $c_1 \neq 0$ ,  $Ac_1 = \lambda_1 c_1$ . Complete  $c_1$  to a basis for  $\mathbb{F}^n$ :

$$\Rightarrow A \underbrace{[c_1, \tilde{c}_2, \dots, \tilde{c}_n]}_E = \underbrace{[c_1, \tilde{c}_2, \dots, \tilde{c}_n]}_E \begin{bmatrix} \lambda_1 & * \cdots * \\ 0 & C \end{bmatrix}.$$

Now, eigenvalues of  $C$  must be also eigenvalues of  $A$  (because  $E^{-1}AE = \begin{pmatrix} \lambda_1 & * \\ 0 & C \end{pmatrix}$ ).

So, by induction, there exists an invertible  $D$  such that  $D^{-1}CD$  is triangular with eigenvalues appearing along the diagonal the way we want them to be. But then

$$\underbrace{AE}_{B} \begin{bmatrix} 1 & 0 \\ 0 & D \end{bmatrix} = \underbrace{E}_{B} \begin{bmatrix} 1 & 0 \\ 0 & D \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & D^{-1} \end{bmatrix} \begin{bmatrix} \lambda_1 & * \\ 0 & C \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & D \end{bmatrix}}_R$$

and  $R$  is upper triangular as desired (note that  $B$  is invertible since it is the product of invertible matrices).  $\square$

**Example 1.5.38** *We must stress that the transformation to triangular form  $R$  is not unique. What is unique are the entries on the diagonal of  $R$ , but not the off-*

*diagonal entries. For example, take  $A = \begin{bmatrix} 1 & 2 \\ -2 & -3 \end{bmatrix} \in \mathbb{Q}^{2 \times 2}$ . We have  $\lambda_1 = \lambda_2 = -1$ ,*

*and with  $V = \begin{bmatrix} 1 & 0 \\ -1 & 1/2 \end{bmatrix}$  we have  $V^{-1}AV = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ . At the same time, by taking*

*$W = V \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$  we obtain that  $W^{-1}AW = \begin{bmatrix} -1 & 3/2 \\ 0 & -1 \end{bmatrix}$ .*

**Corollary 1.5.39** *Let  $T \in \text{Hom}(\mathbb{F}^n, \mathbb{F}^n)$  and let  $A \in \mathbb{F}^{n \times n}$  be a matrix representation of  $T$ . If all eigenvalues of  $A$  (equivalently, of the transformation  $T$ ) are in  $\mathbb{F}$ , then there exists a monic polynomial  $q(x) \in \mathbb{F}[x]$  of degree  $n$  such that  $q(A) = 0$  (equivalently,  $q(T) = 0$ ).*

**Pf.** Take the basis given by the columns of  $B$  in Theorem 1.5.37, call these  $v_1, \dots, v_n$ :

$$A[v_1, \dots, v_n] = [v_1, \dots, v_n] \begin{bmatrix} \lambda_1 & r_{12} & \cdots & r_{1n} \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & r_{n-1,n} \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix} :$$

$$\Rightarrow \begin{cases} Av_1 = \lambda_1 v_1 \\ Av_2 = r_{12}v_1 + \lambda_2 v_2 \\ \vdots \\ Av_i = r_{1i}v_1 + r_{2i}v_2 + \cdots + r_{i-1,i}v_{i-1} + \lambda_i v_i, \quad i = 1, 2, \dots, n \end{cases}$$

$$\Rightarrow \begin{cases} (A - \lambda_1 I)v_1 = 0 \\ (A - \lambda_2 I)v_2 = r_{12}v_1 \\ \vdots \\ (A - \lambda_i I)v_i = r_{1i}v_1 + r_{2i}v_2 + \cdots + r_{i-1,i}v_{i-1}, \quad i = 1, 2, \dots, n. \end{cases}$$

Now, observe that

$$(A - \lambda_1 I)(A - \lambda_2 I) = (A - \lambda_2 I)(A - \lambda_1 I) \quad \text{etc.,}$$

$$\therefore (A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_i I)v_k = 0, \quad k = 1, \dots, i$$

$$\therefore \underbrace{(A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_n I)}_S v_k = 0, \quad k = 1, \dots, n$$

$$\therefore S \text{ annihilates a basis of } V \therefore S \equiv 0$$

$$\therefore A \text{ satisfies the polynomial equation } (x - \lambda_1) \cdots (x - \lambda_n) = 0. \quad \square$$

**Remarks 1.5.40**

- (1) *It is worth stressing that to get triangular form, we did not need to assume distinct eigenvalues, though we still assumed all eigenvalues in  $\mathbb{F}$ . In particular, from Theorem 1.5.37 and Exercise 1.5.34-(2), we confirm that  $A$  has at most  $n$  eigenvalues (possibly repeated) in  $\mathbb{F}$ .*
- (2) *The polynomial in Corollary 1.5.39 is effectively the characteristic polynomial. Corollary 1.5.39, then, is the celebrated Cayley–Hamilton theorem.*

- (3) A consequence of the fact that (under the assumptions there) the polynomial  $q(\cdot)$  in Corollary 1.5.39 is the characteristic polynomial, is that the minimal polynomial of  $T$  cannot have degree higher than  $n$ ; cfr. with Theorem 1.3.11.

**Exercise 1.5.41** Let  $A, B \in \mathbb{F}^{n \times n}$  have all eigenvalues in  $\mathbb{F}$ . Show that if  $AB = BA$ , then  $A$  and  $B$  are simultaneously triangularizable over  $\mathbb{F}$  (i.e., there exists an invertible  $V \in \mathbb{F}^{n \times n}$  such that  $V^{-1}AV$  and  $V^{-1}BV$  are both triangular). You can assume that  $A$  and  $B$  have a common eigenvector, a fact that will be proven in Theorem 1.5.66 below. [Hint. Proceed by induction on  $n$ .]

By providing a counterexample, show that the reverse implication is not true.

- Our next step will be to further simplify, if possible, the triangular structure we achieved. (Remember that we can get diagonal form over  $\mathbb{F}$  if all eigenvalues are distinct and in  $\mathbb{F}$ ). The key simplification step will be developed in Exercise 1.5.42 below.

**Exercise 1.5.42** We are going to put together several of the previous ingredients.

Given  $R \in \mathbb{F}^{(n+m) \times (n+m)}$ ,  $R = \begin{pmatrix} A & -C \\ 0 & B \end{pmatrix}$  where  $C \in \mathbb{F}^{n \times m}$ , and

$$A \in \mathbb{F}^{n \times n} = \begin{pmatrix} \lambda & a_{12} & \cdots & a_{1n} \\ 0 & \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ 0 & \cdots & 0 & \lambda \end{pmatrix}, \quad B \in \mathbb{F}^{m \times m} = \begin{pmatrix} \mu & b_{12} & \cdots & b_{1m} \\ 0 & \mu & \ddots & \vdots \\ \vdots & \ddots & \ddots & b_{m-1,m} \\ 0 & \cdots & 0 & \mu \end{pmatrix}, \quad \lambda \neq \mu.$$

Then, there exists unique  $X \in \mathbb{F}^{n \times m}$  such that  $\begin{pmatrix} I_n & -X \\ 0 & I_m \end{pmatrix} \begin{pmatrix} A & -C \\ 0 & B \end{pmatrix} \begin{pmatrix} I_n & X \\ 0 & I_m \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ .

That is,  $V^{-1}RV = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$  with  $V = \begin{pmatrix} I_n & X \\ 0 & I_m \end{pmatrix}$ .

**Solution:** Note that  $X$  must solve

$$AX - XB = C.$$

We will effectively solve this matrix equation by backward substitution, finding one row of  $X$  at the time, starting from the last row.

Rewrite this matrix equation as

$$\begin{pmatrix} \lambda & a_{12} & \cdots & a_{1n} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \lambda & a_{n-1,n} \\ 0 & \cdots & 0 & \lambda \end{pmatrix} \begin{pmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nm} \end{pmatrix} - \begin{pmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nm} \end{pmatrix} \begin{pmatrix} \mu & b_{12} & \cdots & b_{1m} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \mu & b_{m-1,m} \\ 0 & \cdots & 0 & \mu \end{pmatrix}$$

$$= \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & & \vdots \\ c_{n1} & \cdots & c_{nm} \end{pmatrix}$$

Now solve for the entries of  $X$  from the bottom to the top, left to right.

From  $n$ -th row of  $X$

$$\text{1st column : } \lambda x_{n1} - \mu x_{n1} = c_{n1} \rightarrow x_{n1} = c_{n1}/(\lambda - \mu)$$

$$\text{2nd column : } \lambda x_{n2} - x_{n1}b_{12} - x_{n2}\mu = c_{n2} \rightarrow x_{n2} = (c_{n2} + x_{n1}b_{12})/(\lambda - \mu)$$

$$\vdots$$

$$\text{\textit{m}-th column : } \cdots \rightarrow x_{nm}$$

$\Rightarrow n$ -th row of  $X$  found.

Next, we solve for the  $(n-1)$ -st row of  $X$ . 1st column:  $\lambda x_{n-1,1} + a_{n-1,n}x_{n1} - x_{n-1,1}\mu = c_{n-1,1} \rightarrow x_{n-1,1}$ . Keep going.  $\square$

**Exercise 1.5.43** Let  $R \in \mathbb{F}^{(n+m) \times (n+m)}$  be block upper triangular,  $R = \begin{pmatrix} A & -C \\ 0 & B \end{pmatrix}$  where  $C \in \mathbb{F}^{n \times m}$ , and assume that  $R$  has all eigenvalues in  $\mathbb{F}$ , with the eigenvalues of  $A$  and  $B$  being distinct from one another. That is, let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues of  $A$ ,  $\mu_1, \dots, \mu_m$  be the eigenvalues of  $B$ , and assume that  $\lambda_i \neq \mu_j$ , for all  $i = 1, \dots, n$ ,  $j = 1, \dots, m$ . Show that there exists  $V$  such that  $V^{-1}RV = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ . [Hint. Bring  $A$  and  $B$  to triangular form:

$$\begin{pmatrix} \lambda_1 & a_{12} & \cdots & a_{1n} \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}, \quad B \in \mathbb{F}^{m \times m} = \begin{pmatrix} \mu_1 & b_{12} & \cdots & b_{1m} \\ 0 & \mu_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & b_{n-1,n} \\ 0 & \cdots & 0 & \mu_m \end{pmatrix},$$

and use the construction of Exercise 1.5.42.]

**Exercise 1.5.44** Let  $R \in \mathbb{F}^{n \times n}$  be a block upper triangular matrix of the form

$$R = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1p} \\ 0 & R_{22} & \cdots & R_{2p} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & R_{pp} \end{bmatrix},$$

where each block  $R_{jj} \in \mathbb{F}^{n_j \times n_j}$  is upper triangular, with constant diagonal given by  $\lambda_j$ ,  $j = 1, \dots, p$ ,  $n_1 + \dots + n_p = n$ . Assume that  $\lambda_j \neq \lambda_k$  for  $j \neq k$ ,  $j, k = 1, \dots, p$ . Show that there exists a similarity transformation matrix  $V \in \mathbb{F}^{n \times n}$  such that  $V^{-1}RV = \text{diag}(R_{jj}, j = 1, \dots, p)$ . [Hint: There exists, unique,  $V$  of the form

$$V = \begin{bmatrix} I_{n_1} & X_{12} & \cdots & X_{1p} \\ 0 & I_{n_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & I_{n_p} \end{bmatrix}, \quad X_{ij} \in \mathbb{F}^{n_i \times n_j}, \quad \text{such that } V^{-1}RV = \begin{bmatrix} R_{11} & 0 & \cdots & 0 \\ 0 & R_{12} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & R_{pp} \end{bmatrix}.$$

As a consequence of Exercises 1.5.43 and 1.5.44, we have

**Corollary 1.5.45 (Block Diagonal Form)** If  $A \in \mathbb{F}^{n \times n}$  has all eigenvalues in  $\mathbb{F}$ ,

then there exists  $T \in \mathbb{F}^{n \times n}$ , invertible, such that  $T^{-1}AT$  is block diagonal,  $\begin{pmatrix} R_{11} & 0 & \cdots & 0 \\ 0 & R_{12} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & R_{pp} \end{pmatrix}$ ,

where each  $R_{ii} \in \mathbb{F}^{n_i \times n_i}$ , has eigenvalues  $\lambda_i$ ,  $i = 1, \dots, p$ , and  $\lambda_i \neq \lambda_j$ ,  $i \neq j$ , and  $n_1 + \dots + n_p = n$ . Moreover, we can select each  $R_{ii}$  to be upper triangular with

constant diagonal:  $R_{ii} = \begin{pmatrix} \lambda_i & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_i \end{pmatrix}$ , and  $\lambda_i \neq \lambda_j$ ,  $i \neq j$ ,  $i, j = 1, \dots, p$ .

**Exercise 1.5.46** Show Theorem 1.5.47 below.

**Theorem 1.5.47 (On the degree of uniqueness of block diagonal reduction)**

Let  $A \in \mathbb{F}^{n \times n}$  with all eigenvalues in  $\mathbb{F}$ , let invertible  $T \in \mathbb{F}^{n \times n}$  be such that  $T^{-1}AT$  is block diagonal:

$$T^{-1}AT = B = \begin{pmatrix} B_{11} & 0 & \cdots & 0 \\ 0 & B_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & B_{pp} \end{pmatrix}, \quad \text{where } B_{ii} \in \mathbb{F}^{n_i \times n_i}, \quad n_1 + \dots + n_p = n, \quad \text{and}$$

the eigenvalues of  $B_{ii}$  and of  $B_{jj}$  are disjoint, for  $i \neq j$ . Then, within the class of transformations  $T$  bringing  $A$  to block diagonal form with diagonal blocks having the same eigenvalues as the  $B_{ii}$ 's, the transformation  $T$  is unique up to an invertible block diagonal transformation, that is up to transformations of the type  $T \leftarrow TV$ ,

where  $V = \begin{pmatrix} V_{11} & 0 & \cdots & 0 \\ 0 & V_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & V_{pp} \end{pmatrix}$ , where  $V_{ii} \in \mathbb{F}^{n_i \times n_i}$ ,  $i = 1, \dots, p$ , are invertible, and  $n_1 + \cdots + n_p = n$ .

- Is the form of Corollary 1.5.45, block-diagonal with triangular diagonal blocks, as “simple” a form as we can get? The answer has a lot to do with multiple eigenvalues and nilpotent matrices. To appreciate this fact, let us first give the desired target result, then we’ll see how to get it.

### 1.5.3 Jordan Form

**Definition 1.5.48** A matrix  $J_p(\lambda) \in \mathbb{F}^{p \times p}$  is called a Jordan block of size  $p$ , associated to the eigenvalue  $\lambda$ , if it is of the form  $J_p(\lambda) =$

$$J_p(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}. \quad A$$

matrix  $J \in \mathbb{F}^{n \times n}$  is said to be in Jordan form if it is the direct sum of Jordan blocks:

$$J = \begin{pmatrix} J_{n_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{n_2}(\lambda_2) & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & J_{n_k}(\lambda_k) \end{pmatrix}, \text{ where } n_1 + n_2 + \cdots + n_k = n. \text{ The } \lambda_i \text{'s need not be distinct.}$$

#### Remarks 1.5.49

- (1) If  $k = n$  (hence  $n_i = 1$ , for all  $i$ ), then  $J$  is diagonal.
- (2) Relative to the same eigenvalue, customarily one chooses (as we will do below) the sizes of the Jordan blocks in decreasing order.
- (3) If  $k < n$ , then  $J$  cannot be transformed to diagonal form.  
(If  $k < n \Rightarrow$  there is one block, say  $J_p \in \mathbb{F}^{p \times p}$ , with  $p > 1$ . We show that this block cannot be diagonalized. For, suppose the contrary:  $\exists S : S^{-1}J_p S = \Lambda_p$ , diagonal.

$$\text{Then } \Lambda_p = \begin{pmatrix} \lambda_p & 0 & \cdots & 0 \\ 0 & \lambda_p & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_p \end{pmatrix} = \lambda_p I \Rightarrow S^{-1}J_p S = \lambda_p I \Rightarrow J_p = \lambda_p I, \text{ which cannot be true if } p > 1.)$$

(4) Obviously, there is only an “eigendirection” associated to  $\begin{pmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \lambda & 1 \\ & & & \lambda \end{pmatrix}$ . The

only eigenvector must be a nonzero multiple of  $\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ .

(5) A Jordan block is a special type of a (upper) bidiagonal matrix, where a matrix  $A \in \mathbb{F}^{n \times n}$  is said to be upper bidiagonal when  $a_{i,j} = 0$  for  $i > j$  and  $j > i + 1$ ,  $i, j = 1, \dots, n$ .

We are ready for the **important** result that matrices in  $\mathbb{F}^{n \times n}$  with eigenvalues in  $\mathbb{F}$  can be brought to Jordan form. In light of Remarks 1.5.49-(3)-(4) above, this form cannot be further simplified; that is, the form is as close as possible to being diagonal.

**Theorem 1.5.50 (Jordan canonical form)** *Given  $A \in \mathbb{F}^{n \times n}$  with all eigenvalues in  $\mathbb{F}$ . Then, there exists  $S \in \mathbb{F}^{n \times n}$ , invertible, such that  $S^{-1}AS$  is in Jordan form. This form (that is, the values  $n_1, \dots, n_k$ ) is unique, aside from trivial reordering of the diagonal blocks.*

To prove this result we make some preliminary observations, which will simplify our life.

### Observations 1.5.51

(1) Because of Corollary 1.5.45, we can assume – without loss of generality – that the matrix  $A$  is in block diagonal form and the diagonal blocks are upper triangular, each corresponding to a different eigenvalue. Clearly, if we can bring to their respective Jordan forms all of these blocks on the diagonal of  $A$ , then we can bring  $A$  to Jordan form.

(2) So, we may as well assume that  $A \in \mathbb{F}^{n \times n}$  is given by a single triangular block:  $A = \lambda I + N$ , where  $N$  is the strictly upper triangular part of  $A$ . That is,  $A =$

$$\begin{pmatrix} \lambda & * & \cdots & * \\ 0 & \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda \end{pmatrix} = \lambda I + N. \text{ Since } S^{-1}AS = \lambda I + S^{-1}NS, \text{ we will only need to prove}$$

that  $N$  can be taken to Jordan form as claimed. So, we will focus on transforming

$N \in \mathbb{F}^{n \times n}$  to Jordan form, where

$$N = \begin{pmatrix} 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & & * \\ & & & 0 \end{pmatrix}. \quad (1.5.7)$$

Observe that  $N$  is nilpotent and for sure  $N^n = 0$ .

**Definition 1.5.52** A given nilpotent matrix  $N \in \mathbb{F}^{n \times n}$  is said to have index of nilpotency  $n_1$ , if  $N^{n_1} = 0$  but  $N^{n_1-1} \neq 0$ .<sup>5</sup>

**Example 1.5.53** Take

$$N = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}.$$

Easily, we always have  $N^3 = 0$ , but  $N^2 = \begin{pmatrix} 0 & 0 & ab \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , i.e.,  $N^2 = 0$  if and only if  $a$  or  $b$  is 0.

$\therefore$   $N$  has index of nilpotency 3 if  $a \neq 0$ ,  $b \neq 0$ , nilpotency index 2 if  $a = 0$  or  $b = 0$ , and the nilpotency index is 1 if  $a = b = c = 0$ .

In light of Observations 1.5.51, Theorem 1.5.50 will follow from the next one:

**Theorem 1.5.54** Let  $N \in \mathbb{F}^{n \times n}$  be strictly upper triangular as in (1.5.7). Then, there is an invertible  $V \in \mathbb{F}^{n \times n}$  and indices  $n_1, n_2, \dots, n_p : n_1 \geq n_2 \geq \dots \geq n_p \geq 1$ ,  $n_1 + n_2 + \dots + n_p = n$  such that

$$V^{-1}NV = \begin{bmatrix} J_{n_1}(0) & 0 & \cdots & 0 \\ 0 & J_{n_2}(0) & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & J_{n_p}(0) \end{bmatrix}.$$

Here,  $n_1$  is the nilpotency index of  $N$ . The indices  $n_1, \dots, n_p$  are unique.

---

<sup>5</sup>By convention, the 0-th power of any matrix is the identity.

**Example 1.5.55** Take  $N = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}$  and let us examine different cases.

1)  $a = b = c = 0 \Rightarrow N = 0$  and  $n_1 = n_2 = n_3 = 1$ .

2)  $a = b = 0, c \neq 0 \Rightarrow N = \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, N^2 = 0$ .

Observe that  $N$  is similar to  $\tilde{N} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \left( \text{use } \begin{pmatrix} 1/c & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} N \begin{pmatrix} c & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right),$

which is

similar to  $\left( \begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ - & - & - \\ 0 & 0 & 0 \end{array} \right) \left( \text{use } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tilde{N} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \right), \text{ then } n_1 = 2, n_2 = 1.$

3)  $a = 0, b \neq 0, N = \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow N^2 = 0$ .

$N$  is similar to  $\begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \left( \text{use } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & 1 \end{pmatrix} \right), \text{ which is similar to } \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \left( \text{use } \begin{pmatrix} 1 & -c & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right), \text{ which is similar to } \left( \begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ - & - & - \\ 0 & 0 & 0 \end{array} \right) \left( \text{use } \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \right), \text{ so } n_1 = 2,$

$n_2 = 1$ .

4)  $a \neq 0, b = 0 \Rightarrow \text{as in 3}).$

5)  $a \neq 0, b \neq 0, N^2 \neq 0, N^3 = 0; N = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}, \text{ which is similar to}$

$\begin{pmatrix} 0 & ab & c \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \left( \text{use } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & 1 \end{pmatrix} \right), \text{ which is similar to}$

$\begin{pmatrix} 0 & 1 & c/ab \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \left( \text{use } \begin{pmatrix} 1/ab & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right), \text{ which is finally similar to}$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \left( use \begin{pmatrix} 1 & -c/ab & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right), \text{ that is } n_1 = 3. \quad \square$$

**Proof of Theorem 1.5.54.**<sup>6</sup>

By induction on  $n$ . For  $n = 1$ , then  $N = [0]$ , and the statement is trivially true.

So, suppose that the result is true for all strictly upper triangular matrices of size  $< n$ .

$$\text{Write } N = \begin{matrix} & 1 & n-1 \\ & 1 & a^T \\ n-1 & 0 & N_1 \end{matrix} \quad (\text{we write } a^T \text{ to clarify that it is a "row vector"}).$$

$$\text{By induction, } \exists V_1 : V_1^{-1} N V_1 = \begin{bmatrix} J_{k_1} & 0 & \cdots & 0 \\ 0 & J_{k_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & J_{k_s} \end{bmatrix} =: \begin{bmatrix} J_{k_1} & 0 \\ 0 & J \end{bmatrix} \text{ with } k_1 \geq \cdots \geq k_s \geq 1,$$

$k_1 + \cdots + k_s = n - 1$ , and note that  $J^{k_1} = 0$ . So

$$\begin{pmatrix} 1 & 0 \\ 0 & V_1^{-1} \end{pmatrix} N \begin{pmatrix} 1 & 0 \\ 0 & V_1 \end{pmatrix} =: \begin{matrix} & 1 & k_1 & k_2 + \cdots + k_s \\ & 1 & b_1^T & b_2^T \\ k_1 & 0 & J_{k_1} & 0 \\ k_2 + \cdots + k_s & 0 & 0 & J \end{matrix}.$$

This is further transformed by similarity as follows

$$\begin{aligned} & \begin{pmatrix} 1 & -b_1^T J_{k_1}^T & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} 0 & b_1^T & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix} \begin{pmatrix} 1 & b_1^T J_{k_1}^T & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} \\ &= \begin{pmatrix} 0 & b_1^T (I - J_{k_1}^T J_{k_1}) & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix} = \begin{pmatrix} 0 & (b_1^T e_1) e_1^T & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix}, \end{aligned}$$

since (verify!)

$$I - J_{k_1}^T J_{k_1} = e_1 e_1^T \quad \left( e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{F}^{k_1}, \quad \text{with } e_2, e_3, \dots \text{ similarly defined} \right).$$

---

<sup>6</sup>This proof is adapted from [5] and it uses induction and elementary similarity transformations. For an alternative proof, more abstract in nature, see [3].

There are two cases to consider:  $b_1^T e_1 = 0$  and  $b_1^T e_1 \neq 0$ .

1)  $b_1^T e_1 = 0$ . Then, we have  $\begin{pmatrix} 0 & 0 & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix}$  which is similar (via permutation) to

$$\left( \begin{array}{c|cc} J_{k_1} & 0 & 0 \\ \hline - & - & - \\ 0 & 0 & b_2^T \\ 0 & 0 & J \end{array} \right), \text{ but by induction there is } V_2 \in \mathbb{F}^{(n-k_1) \times (n-k_1)} \text{ such that } \tilde{J} = V_2^{-1} \begin{pmatrix} 0 & b_2^T \\ 0 & J \end{pmatrix} V_2 \text{ is in Jordan form.}$$

$\therefore N$  is similar to  $\begin{pmatrix} J_{k_1} & 0 \\ 0 & \tilde{J} \end{pmatrix}$  which is in Jordan form (it may be that blocks are not arranged in non-increasing order, in which case we can permute them to obtain the desired ordering).

2)  $b_1^T e_1 \neq 0$ . Then, we have  $\begin{pmatrix} 0 & [* , 0, \dots, 0] & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix}$ . Now, first we make the “\*” entry become 1:

$$\begin{pmatrix} \frac{1}{b_1^T e_1} & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} 0 & (b_1^T e_1) e_1^T & b_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix} \begin{pmatrix} (b_1^T e_1) & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} =: \begin{pmatrix} 0 & e_1^T & c_2^T \\ 0 & J_{k_1} & 0 \\ 0 & 0 & J \end{pmatrix}$$

Note that  $\widehat{J} := \begin{pmatrix} 0 & e_1^T \\ 0 & J_{k_1} \end{pmatrix}$  is a Jordan block of size  $k_1 + 1$ :  $\widehat{J} = J_{k_1+1}(0)$ . So, we have

$$\begin{matrix} & k_1 + 1 & k_2 + \dots + k_s \\ k_1 + 1 & \begin{pmatrix} \widehat{J} & e_1 c_2^T \\ 0 & J \end{pmatrix} \\ k_2 + \dots + k_s & \end{matrix}$$

Next, we “chase away”  $e_1 c_2^T$  by a sequence of similarities:

$$\begin{pmatrix} I & e_2 c_2^T \\ 0 & I \end{pmatrix} \begin{pmatrix} \widehat{J} & e_1 c_2^T \\ 0 & J \end{pmatrix} \begin{pmatrix} I & -e_2 c_2^T \\ 0 & I \end{pmatrix} = \begin{pmatrix} \widehat{J} & -\widehat{J} e_2 c_2^T + e_1 c_2^T + e_2 c_2^T J \\ 0 & J \end{pmatrix} = \begin{pmatrix} \widehat{J} & e_2 c_2^T J \\ 0 & J \end{pmatrix},$$

since  $\widehat{J} e_2 = e_1$ . And, recursively, for  $i = 1, 2, 3, \dots, k_1$ :

$$\begin{pmatrix} I & e_{i+1} c_2^T J^{i-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \widehat{J} & e_i c_2^T J^{i-1} \\ 0 & J \end{pmatrix} \begin{pmatrix} I & -e_{i+1} c_2^T J^{i-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} \widehat{J} & -\widehat{J}e_{i+1}c_2^T J^{i-1} + e_i c_2^T J^{i-1} + e_{i+1} c_2^T J^i \\ 0 & J \end{pmatrix} = \begin{pmatrix} \widehat{J} & e_{i+1} c_2^T J^i \\ 0 & J \end{pmatrix},$$

since  $\widehat{J}e_{i+1} = e_i$ . But  $J^{k_1} = 0$ , and so we eventually get the form

$$\begin{pmatrix} \widehat{J} & 0 \\ 0 & J \end{pmatrix} = \begin{pmatrix} J_{k_1+1}(0) & 0 & \cdots & 0 \\ 0 & J_{k_2}(0) & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & J_{k_s}(0) \end{pmatrix}$$

as desired. (Here,  $n_1 = k_1 + 1$ ,  $n_j = k_j$ ,  $j = 2, \dots, s$  and  $s = p$ .)

The last thing that we need to justify is the uniqueness part. We have obtained the Jordan form for  $N$ :

$$J = \begin{pmatrix} J_{n_1}(0) & & \\ & \ddots & \\ & & J_{n_p}(0) \end{pmatrix}; \quad n_1 + \cdots + n_p = n, n_1 \geq \cdots \geq n_p \geq 1.$$

We show that the set of Jordan blocks (i.e., the values  $n_1, \dots, n_p$ ) is completely specified by the values  $\text{rank}(J^m)$ ,  $m = 1, 2, \dots, n$ . [This will give the result, because since  $N$  and  $J$  are similar then  $N^m$  and  $J^m$  are too and, thus, their ranks are the same. In other words, the sizes of the Jordan blocks are uniquely specified by  $N$ .]

Now:  $J_{n_k}^m = 0$  if  $m \geq n_k$  and  $\text{rank}(J_{n_k}^{m-1}) - \text{rank}(J_{n_k}^m) = 1$  if  $m \leq n_k$ . Set  $r_m = \text{rank}(J^m)$ , and  $d_m = r_{m-1} - r_m$ ,  $m = 1, 2, \dots$ , with  $r_0 = n$ .

$\therefore d_m$  is the number of Jordan blocks in  $J$  of size  $k \geq m$  and surely  $d_m = 0$ , if  $m > n$ .  
 $\therefore$  the number of Jordan blocks in  $J$  of exact size  $k = m$  is given by  $d_m - d_{m+1} = r_{m-1} - 2r_m + r_{m+1}$ ,  $m = 1, 2, \dots, n$ , and so the number of Jordan blocks of size  $k = m$  is given by  $\text{rank}(J^{m-1}) - 2 \text{rank}(J^m) + \text{rank}(J^{m+1}) = \text{rank}(N^{m-1}) - 2 \text{rank}(N^m) + \text{rank}(N^{m+1})$ .  $\square$

### Remarks 1.5.56

- (1) *The Jordan form is a very important theoretical result, which further allows one to create equivalence classes for matrices with the same Jordan form, in the sense that (relative to matrices with eigenvalues in  $\mathbb{F}$ ) we have:*

**Corollary 1.5.57** *Two matrices  $A, B \in \mathbb{F}^{n \times n}$ , with eigenvalues in  $\mathbb{F}$ , are similar if and only if they have the same Jordan form.*

- (2) *By virtue of Remarks 1.5.49-(3, 4), the number of Jordan blocks corresponding to the same eigenvalue gives the geometric multiplicity of that eigenvalue. The*

algebraic multiplicity of that eigenvalue is the sum of the sizes of all Jordan blocks relative to that eigenvalue. [As an aside observation, this fact tells us that the algebraic multiplicity of an eigenvalue cannot be less than its geometric multiplicity.]

- (3) Note that the uniqueness part pertains to the size of the Jordan blocks, not to the transformation giving the Jordan form. For example, take  $A = \begin{bmatrix} 1 & 2 \\ -2 & -3 \end{bmatrix}$ ,  $V = \begin{bmatrix} 1 & 0 \\ -1 & 1/2 \end{bmatrix}$  and  $T = \begin{bmatrix} 1 & 2 \\ -1 & -3/2 \end{bmatrix}$ . Then, we obtain  $V^{-1}AV = T^{-1}AT = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ .

**Exercise 1.5.58** Verify Corollary 1.5.57.

**Exercise 1.5.59** Prove the following statement, or provide a counterexample to it. “If  $A \in \mathbb{F}^{n \times n}$  is nilpotent of index  $n_A$ , and  $B \in \mathbb{F}^{n \times n}$  is nilpotent of index  $n_B$ , then  $(A + B)^{n_A + n_B} = 0$ .”

## 1.5.4 Extending the field

To obtain the Jordan form of a matrix  $A \in \mathbb{F}^{n \times n}$  (over an arbitrary field  $\mathbb{F}$ ), we have assumed that all the eigenvalues are in  $\mathbb{F}$ . This restriction cannot be removed.

So, to proceed further, we need to understand where can be the eigenvalues of a matrix in  $\mathbb{F}^{n \times n}$ , if not in  $\mathbb{F}$ . Or, alternatively, can we have some situations (that is, some special classes of transformations/matrices, and/or of  $\mathbb{F}$ ) which are guaranteed to give eigenvalues in  $\mathbb{F}$ ?

The idea is as follows. We have  $A \in \mathbb{F}^{n \times n}$ , but we can also think of  $A \in \mathbb{K}^{n \times n}$ , where  $\mathbb{K}$  is an *extension* of the field  $\mathbb{F}$ . And, we will want to think of the extension  $\mathbb{K}$  so that **all** the **eigenvalues** of  $A$  will be in  $\mathbb{K}$ . That is, so that all the values  $\lambda$  such that

$$A - \lambda I \text{ is singular} \quad (\text{in } \mathbb{K}^{n \times n}),$$

will be in  $\mathbb{K}$ .

To decide what  $\mathbb{K}$  should be, we recall that the eigenvalues must be roots of the characteristic polynomial, since  $A - \lambda I$  singular  $\Leftrightarrow \det(A - \lambda I) = 0$  and as we know  $\det(\lambda I - A)$  is a polynomial in  $\lambda$  of degree  $n$  with coefficients in  $\mathbb{F}$ . Therefore, the extension field  $\mathbb{K}$  should be chosen so to include all the roots of the characteristic polynomial, that is of all monic algebraic equations of degree  $n$  with coefficients in  $\mathbb{F}$ . In general (that is, unless we are willing to restrict to special classes of matrices),

this practically forces us to consider the complex field  $\mathbb{C}$ , which is a *closed* field<sup>7</sup> and this will allow us to deal with all commonly used numerical fields  $(\mathbb{Q}, \mathbb{R}, \mathbb{C})$ .

**Exercises 1.5.60** Let  $A \in \mathbb{F}^{n \times n}$ .

- (1) Show that  $\det A = \prod_{j=1}^n \lambda_j$  and that  $(-1)^n \det A$  is the constant term in the characteristic polynomial. In particular, this shows that  $A$  is singular if and only if  $\det A = 0$ . Moreover, this also shows that  $\prod_{j=1}^n \lambda_j \in \mathbb{F}$  even if some (or all) of the  $\lambda_j$ 's are in  $\mathbb{K}$  whereas  $A \in \mathbb{F}^{n \times n}$  ( $\mathbb{K}$  being the extension field of  $\mathbb{F}$ ).
- (2) Show that if  $A$  is invertible, then the eigenvalues of  $A^{-1}$  are the reciprocal of the eigenvalues of  $A$ . In particular, this confirms that  $\det A^{-1} = 1/\det A$ .
- (3) Let the trace of  $A$  be defined as  $\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$ . By using (1.5.1), show that  $-\text{tr}(A)$  is the coefficient relative to degree  $(n-1)$  in the characteristic polynomial (1.5.5). Moreover, show that  $\text{tr}(A) = \sum_{j=1}^n \lambda_j$ . (This also shows that  $\sum_{j=1}^n \lambda_j \in \mathbb{F}$  even if some or all of the  $\lambda_j$ 's are not in  $\mathbb{F}$ ).

**Remark 1.5.61** The above Exercises imply that  $\det A$  and  $\text{tr} A$  are invariant under similarity transformations.

In particular, from Theorem 1.5.50 and closeness of the field of complex numbers, we have proven the following fundamental result.

**Theorem 1.5.62** Any matrix  $A \in \mathbb{C}^{n \times n}$  is similar to a Jordan form as in Definition 1.5.48. That is, for any given  $A \in \mathbb{C}^{n \times n}$ , there exists an invertible  $V \in \mathbb{C}^{n \times n}$  such that  $V^{-1}AV$  is in Jordan form.  $\square$

**Exercise 1.5.63** Show that if  $T \in \text{Hom}(\mathbb{C}^n, \mathbb{C}^n)$  is nilpotent (see Exercise 1.3.13-(6)), that is  $T^k = 0$  for some  $k$ , then  $T^n = 0$ . Show that if  $A \in \mathbb{C}^{n \times n}$  is any matrix representation of  $T$ , then all eigenvalues of  $A$  are 0. Is the result true also if  $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ ?

As a consequence of Theorem 1.5.62, one may as well assume (from the theoretical point of view) that a given matrix  $A \in \mathbb{C}^{n \times n}$  is in Jordan form, in that it can be brought to Jordan form if it is not so to begin with. This line of thought allows us to prove interesting facts in a simple way, as exemplified by the next exercise.

**Exercise 1.5.64** Let  $A \in \mathbb{C}^{n \times n}$ . Suppose there exists an integer  $k > 1$  such that  $A^k = A$ . Show that  $A$  is diagonalizable.

---

<sup>7</sup>A field  $\mathbb{F}$  is closed if all roots of polynomial equations with coefficients in  $\mathbb{F}$  are also in  $\mathbb{F}$ .

**Solution.** Without loss of generality, we can assume that  $A$  is in Jordan form

$$J = \text{diag}(J_{n_k}(\lambda_k), k = 1, \dots, p).$$

By contradiction, we suppose that at least one Jordan block is non-diagonal (i.e., at least one  $n_k > 1$ , for some  $k = 1, \dots, p$ ). Since  $A^k = A$  implies  $J^k = J$ , and  $J$  is block diagonal, we can assume that we are dealing with just one Jordan block:

$$J = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ & & & \ddots & 1 \\ & & & & \lambda \end{pmatrix}. \text{ It is immediate to realize (use induction) that } J^k \text{ has the}$$

$$\text{form } J^k = \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \dots & \dots \\ & \lambda^k & \ddots & \\ & & \ddots & \ddots \\ & & & \ddots & k\lambda^{k-1} \\ & & & & \lambda^k \end{pmatrix}. \text{ Note that if } \lambda = 0, \text{ then it is clearly not}$$

possible to have  $J^k = J$  for  $k > 1$ . Now, to have  $J^k = J$ , we need to have

$$(a) \quad \lambda^k = \lambda \quad \text{and} \quad (b) \quad k\lambda^{k-1} = 1.$$

From (a), we have either  $\lambda = 0$  or  $\lambda^{k-1} = 1$ . As above, we cannot have  $\lambda = 0$ , so using (b) we need  $k = 1$ , which is a contradiction. Therefore, all Jordan blocks must be diagonal.  $\square$

**Exercises 1.5.65** Here, we explore the connection between Jordan form and minimal and characteristic polynomials. You may assume that  $A \in \mathbb{C}^{n \times n}$  or that  $A \in \mathbb{F}^{n \times n}$  with all eigenvalues in  $\mathbb{F}$ .

(1) Suppose that the distinct eigenvalues of  $A$  are  $\lambda_1, \dots, \lambda_k$ . Show that the minimal polynomial of  $A$  is  $p(x) = \prod_{j=1}^k (x - \lambda_j)^{n_j}$ , where  $n_j$  is the size of the largest Jordan block relative to  $\lambda_j$ .<sup>8</sup> Conclude that  $A$  is diagonalizable if and only if its minimal polynomial is simply  $(x - \lambda_1) \cdots (x - \lambda_k)$ . What is the characteristic polynomial of  $A$ ?

(2) Consider the following matrix  $C$ :

$$C = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & 1 & 0 & -a_{n-2} \\ 0 & \dots & 0 & 1 & -a_{n-1} \end{pmatrix}.$$

---

<sup>8</sup>How are the  $n_j$ 's related to the nilpotency index?

Show that the characteristic and minimal polynomials of  $C$  are the same. [ $C$  is called the companion matrix of its characteristic polynomial.] Note that we are not saying that a certain matrix  $A$  with given characteristic polynomial  $p(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + x^n$  is similar to  $C$ ; however, show that  $A$  is similar to  $C$  if and only if the minimal and characteristic polynomials of  $A$  are the same.

### Commuting Matrices.

Next, we fill a gap left open in Exercise 1.5.41, about commuting matrices sharing an eigenvector. To prove the next result, we will make use of Lemma 1.5.35 and we give the result over an arbitrary field  $\mathbb{F}$ , though we can think of  $\mathbb{F} = \mathbb{C}$ .

**Theorem 1.5.66** *Let  $A, B \in \mathbb{F}^{n \times n}$  with all eigenvalues in  $\mathbb{F}$ , and let  $AB = BA$ . Then,  $A$  and  $B$  share an eigenvector.*

**Pf.** The first step of our proof is to simplify the structure of the problem.

Let  $V \in \mathbb{F}^{n \times n}$ , invertible, be such that  $V^{-1}AV$  is in block-diagonal form (see Corollary 1.5.42):

$$V^{-1}AV = \begin{pmatrix} R_{11} & & \\ & \ddots & \\ & & R_{pp} \end{pmatrix},$$

where each  $R_{ii}$  is upper triangular with constant diagonal:  $R_{ii} = \lambda_i I_{n_i} + N_i$ ,  $\lambda_i \neq \lambda_j$ ,  $i \neq j$ ,  $i, j = 1, \dots, p$ , and  $N_i$ 's are strictly upper triangular.

Now, the relation  $AB = BA$  can be rewritten as

$$(V^{-1}AV)(V^{-1}BV) = (V^{-1}BV)(V^{-1}AV)$$

and from this, partitioning  $V^{-1}BV$  as  $V^{-1}AV$  is, and using that  $\lambda_i \neq \lambda_j$ ,  $i \neq j$ , it is easy to realize that we must have  $(V^{-1}BV)_{ij} = 0$ ,  $i \neq j$ ,  $i, j = 2, \dots, p$ . Therefore, with no loss of generality we can take  $A$  and  $B$  both block-diagonals, with the diagonal blocks of  $A$  of the form  $\lambda_i I + N$ .

In other words, it is enough for us to prove the result for the problem  $A = \lambda I + N$ ,  $N \in \mathbb{F}^{n \times n}$  strictly upper triangular, and  $B \in \mathbb{F}^{n \times n}$  being any matrix commuting with  $A$ . But, then, the relation  $AB = BA$  is the same as  $NB = BN$  where all eigenvalues of  $N$  are 0, so we will show that  $N$  and  $B$  share an eigenvector.

Now, let  $S$  give the Jordan form of  $N$ :  $NS = SJ$ , and without loss of generality assume that  $J$  has  $p$  blocks with  $p < n$  (otherwise, if  $p = n$ ,  $S$  diagonalizes  $N$  and

thus  $N = 0$ , and any eigenvector of  $B$  will do). So:

$$J = \begin{bmatrix} J_1(0) & & \\ & \ddots & \\ & & J_p(0) \end{bmatrix},$$

where each  $J_k(0)$  is a Jordan block of size  $n_k$ ,  $n_1 + \dots + n_p = n$ , and  $p < n$ .

Let  $\mathcal{V}$  be the span of the linearly independent eigenvectors of  $N$ , of which we know that there are exactly  $p$  (see Remark 1.5.56-(2)), call them  $v_1, \dots, v_p$ . Take any vector  $v \in \mathcal{V}$ ,  $v \neq 0$ , and observe that  $Nv = 0 \cdot v = 0$  and therefore  $N(Bv) = B(Nv) = 0 \cdot Bv$ . We have two possibilities.

- (a)  $Bv = 0$ , which means that  $v$  is eigenvector of  $B$  with eigenvalue 0, and we are done. Or
- (b)  $Bv \neq 0$ . But then  $Bv$  is an eigenvector of  $N$ , and thus  $Bv = c_1 v_1 + \dots + c_p v_p$ , with not all  $c_i$ 's equal to 0.

Now, take progressively  $v = v_1, v = v_2, \dots, v = v_p$ , to get

$$B[v_1, \dots, v_p] = [v_1, \dots, v_p] \begin{bmatrix} c_{11} & \dots & c_{1p} \\ & \dots & \\ c_{p1} & \dots & c_{pp} \end{bmatrix} =: [v_1, \dots, v_p]C, \quad C \in \mathbb{F}^{p \times p},$$

and from Lemma 1.5.35 we get that there is  $w \in \mathcal{V}$  such that  $w$  is an eigenvector of  $B$ . Since  $w \in \mathcal{V}$ , it must also satisfy  $Nw = 0$  and so it is also an eigenvector of  $N$ .  $\square$

### Practical considerations on the Jordan form.

Suppose we have a matrix  $A \in \mathbb{C}^{n \times n}$  (or  $A \in \mathbb{F}^{n \times n}$  with eigenvalues in  $\mathbb{F}$ ), and want to know what is its Jordan form. In general, we may think to proceed as follows.

- (1) Find the upper triangular form of Theorem 1.5.37 with equal eigenvalues appearing consecutively along the diagonal. [When  $\mathbb{F} = \mathbb{C}$ , this reduction can in principle be performed “stably”; see Theorem 3.1.1.]
- (2) Find the block diagonal form of Corollary 1.5.45. [“Stability” of this transformation is not at all obvious; see Exercise 1.5.68.]
- (3) For each triangular block of the block-diagonal form of point (2), consider the corresponding nilpotent matrix  $N$ . For this one, find the dimensions of the Jordan blocks as in the last part of the proof of Theorem 1.5.54. Finally, assemble the overall Jordan form  $J$ .

- (4) Once we have the target result,  $J$ , we can finally find a  $V$  which gives us  $V^{-1}AV = J$ . See Exercises 7.1.10 and 7.1.11.

Unfortunately, the practical use of the Jordan form of a matrix is somewhat limited, since computing the Jordan form is a very delicate task and we should be careful about its use in practical (numerical) computations. The difficulty is that the Jordan form itself is not continuous in the entries of the matrix, and small changes in the entries of a matrix can produce Jordan forms with different blocks sizes. In other words, the size of the Jordan blocks is not stable under perturbations.

**Example 1.5.67** Consider the matrix  $A = \begin{pmatrix} 1 & 2 \\ -2 & -3 \end{pmatrix}$ , which has the repeated eigenvalue  $\lambda = -1$  and is not diagonalizable. Its Jordan form is given, for example, by the similarity  $V = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$ :  $V^{-1}AV = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ . Now, for  $\varepsilon > 0$  and small, take the perturbation  $E = \begin{pmatrix} \varepsilon & -\varepsilon \\ -\varepsilon & -\varepsilon \end{pmatrix}$ , and the matrix  $B = A + E$ . Clearly,  $B \rightarrow_{\varepsilon \rightarrow 0} A$ . For each given  $\varepsilon > 0$ ,  $B$  is diagonalizable with eigenvalues  $\lambda_{\pm} = -1 \pm \sqrt{4\varepsilon + 2\varepsilon^2}$  and eigenvectors matrix  $W = \begin{pmatrix} x_+ & x_- \\ 1 & 1 \end{pmatrix} \equiv \begin{pmatrix} -\frac{3+\varepsilon+\lambda_+}{2+\varepsilon} & -\frac{3+\varepsilon+\lambda_-}{2+\varepsilon} \\ 1 & 1 \end{pmatrix}$ . Easily,  $\lambda_{\pm} = -1 \pm 2\sqrt{\varepsilon} + \mathcal{O}(\varepsilon)$ , and  $x_{\pm} = -1 \pm \sqrt{\varepsilon} + \mathcal{O}(\varepsilon)$ . Clearly,  $\lambda_{\pm}$  are small perturbations of  $-1$ , and indeed  $\lambda_{\pm} \rightarrow -1$  as  $\varepsilon \rightarrow 0$ , but the matrix  $W$  has only the first column in common with that of  $V$  (directionally). Also, note that the columns of  $W$  become the same as  $\varepsilon \rightarrow 0$ ; hence, when  $\varepsilon$  is small,  $W$  is nearly singular. Finally, note that the perturbed eigenvalues are  $\sqrt{\varepsilon}$ -perturbation of the unperturbed eigenvalues.

**Exercise 1.5.68** The gist of this exercise is to make us realize that we can stably perturb a coarser block structure than the Jordan structure.

Consider the following block diagonal matrix  $R \in \mathbb{C}^{n \times n}$ ,  $R = \begin{bmatrix} R_1 & 0 \\ 0 & R_2 \end{bmatrix}$  where  $R_i \in \mathbb{C}^{n_i \times n_i}$ ,  $i = 1, 2$ ,  $n_1 + n_2 = n$ , and for  $i = 1, 2$ ,  $R_i$  have the form  $R_i = \begin{bmatrix} \lambda_i & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_i \end{bmatrix}$ ,  $\lambda_1 \neq \lambda_2$ . Consider the class of perturbation matrices of the form  $S = \varepsilon \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ , where  $\varepsilon$  is a small positive value. Argue that the block diagonal form  $R$  is stable with

respect to these perturbations. That is, show that there is a similarity transformation  $T = I + \mathcal{O}(\varepsilon)$  such that  $T^{-1}(R + S)T = \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}$ , where each  $U_i = R_i + \mathcal{O}(\varepsilon)$ ,  $i = 1, 2$ . Find expressions that need to be satisfied by the terms up to the third power of  $\varepsilon$  in the expansion of  $T$  in powers of  $\varepsilon$ . Single out the dependency of the terms in the expansion on the difference between  $\lambda_1$  and  $\lambda_2$ . [Hint: write  $T = \begin{bmatrix} I & X \\ 0 & I \end{bmatrix}$ ,  $X = X_0 + \varepsilon X_1 + \dots$ ]

Based on Exercise 1.5.68, we next would like to argue that the blocks  $U_i$  can be stably triangularized; this we will see in Chapter 3.

**Remark 1.5.69** *In these Lecture Notes, we will not look at other canonical forms, the so-called rational canonical forms (nothing to do with  $\mathbb{Q}$ ), which maintain us in the original field  $\mathbb{F}$  (and not an extension), since they are seldom used. If interested, please refer to [3, 5].*

We conclude this first Chapter with two questions.

**Question 1.** As Example 1.5.67 showed, the Jordan form is not stable with respect to perturbations. Moreover, even when a matrix is diagonalizable, the diagonalizing transformation can become nearly singular, which is clearly undesirable. Is there a reduced form, for example triangular, which can be arrived at with transformations which are guaranteed to be well behaved? Are there matrices which can be diagonalized with transformations which are guaranteed to be well behaved?

**Question 2.** Suppose we have a matrix in  $A \in \mathbb{R}^{n \times n}$ . Suppose we have found its eigenvalues (perhaps, as roots of the characteristic polynomial), and have that some of these are complex conjugate. Surely we can, say, triangularize  $A$  with respect to  $\mathbb{C}$ . But, what can we achieve if we insist that the similarities be real? We will see the most important case in Theorem 3.1.5.

To give more complete answers to the questions above, we will restrict to  $\mathbb{R}^n$  and  $\mathbb{C}^n$ , endow them with inner product structure, and exploit the property of orthogonality.

